Nonlinear Schrödinger Equations and Physics-Informed Learning

Research review and a water-wave Dysthe agenda

Literature cutoff: 25 October 2026

Abstract

Physics-informed neural networks have been used to approximate nonlinear Schrödinger solutions, infer coefficients, reconstruct fields from sparse observations, and learn reusable solution operators. These are different tasks, and progress on one does not establish superiority on the others. This review separates direct NLS evidence, water-wave evidence, transferable training methods, and very recent work with limited accessible evidence. The recommended first project asks whether the approved spatial Dysthe equation improves sparse-observation reconstruction and downstream prediction beyond retuned NLS and classical numerical assimilation. Five concrete research questions, a reproducible benchmark, and a staged plan are provided. No published experiment was independently reproduced for this review.

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1 Recommendation and review method

Recommended first paper. Ask: *When does spatial Dysthe physics improve reconstruction and downstream prediction from sparse water-wave observations, after NLS is allowed to fit effective coefficients?* The contribution should be a validated map of when the additional physics is useful, including uncertainty and computational cost. It should not rest solely on substituting a more complicated residual into a familiar neural network.

The clearest published motivation is Ehlers et al.'s water-wave NLS study, whose Conclusions propose Dysthe as a next model [1]. However, subsequent papers already address fully nonlinear wave reconstruction and reusable physics-informed operators [2, 3]. Therefore, generic claims of the first PINN for water-wave reconstruction or a fast water-wave PINO are unsuitable. The more defensible opportunity is a controlled comparison of physics, identifiability, phase accuracy, and generalization in the specified equation.

What newest and best mean here. This is a targeted critical review of primary literature available by 25 October 2026, not an exhaustive systematic review or a universal ranking. Searches combined NLS/Schrödinger, PINN, water waves, Dysthe, conservation, causal training, optimization, error bounds, sparse assimilation, and operator learning; recent searches emphasized 2024–2026. Publisher records, author manuscripts, proceedings, author code, and data repositories were consulted. Publication status was checked separately from arXiv submission dates. The source ledgers supplied with the report record access and limitations. A search failure is not proof of absence.

Priority is based on relevance to the task, quality of comparisons, independent test data, access to methods and code, and clarity about costs. There is no common benchmark establishing one best NLS–PINN method across forward, inverse, long-distance, and many-query problems. Numerical errors reported on different domains, norms, equations, and data budgets are not a leaderboard. Very recent publisher abstracts are useful leads, but provide less inspectable evidence than a complete manuscript and executable release.

Reading priorities. Read Ehlers et al. (2024) for the direct research motivation; Urbán et al. (2025) for a serious compact NLS training baseline; Zhang et al. (2025) for complex-valued NLS learning; Hao et al. (2026) for hard-constraint ablations; and the 2025 water-wave PINO paper for the many-query comparison [1, 4, 5, 6, 3]. Read the 2026 GRAVIS, Noether, and phase–amplitude

papers as recent method candidates, with the access limitations in Section 4. For a mathematics-led direction, add Alejo et al. and Eiras et al. [7, 8].

Evidence labels. Throughout this report, *reported* means an author's result, not a result reproduced here. *Explicit future work* identifies an actual statement in the inspected source. *Proposed question* is this review's synthesis. An available code repository is evidence of availability, not evidence that the code reproduces every claim. The figures in this report are original analytic illustrations and experiment diagrams, not new PINN experiments.

Scope. The literature survey concerns NLS methods broadly. All proposed active research stays within unidirectional deep-water gravity waves and the one-profile-coordinate spatial free-surface model `water-wave-dysthe-spatial-v1`. General NLS, higher-order NLS, and other Schrödinger demonstrations can motivate algorithms; they are not water-wave Dysthe validation. Changes of formulation, dimension, or coefficients require explicit separate model identities.

2 The PDE and the learning tasks

2.1 What NLS describes

An envelope describes the slowly varying amplitude and phase of an oscillatory wave. For example, a leading-order surface signal has the form

$$\eta(x,t) \approx \mathrm{Re}\{B(x,t)e^{\mathrm{i}(k_0 x-\omega_0 t)}\}.$$

The complex field B contains information about both the height and the position of the carrier oscillations. Matching $|B|$ alone therefore does not ensure accurate surface-wave prediction. Envelope-to-surface conversion, including any retained bound harmonics, must be specified consistently.

A common mathematical normalization of cubic NLS is

$$\mathrm{i}q_s + pq_{yy} + \gamma|q|^2 q = 0, \qquad (s,y) \in [0,S] \times \mathbb{T}_L, \tag{2.1}$$

where q is complex, s is an evolution variable, y is a profile coordinate, $p, \gamma \in \mathbb{R}$, and $\mathbb{T}_L = \mathbb{R}/(L\mathbb{Z})$. Dispersion is represented by pq_{yy} and nonlinear phase modulation by $\gamma|q|^2 q$. Under this convention $p\gamma > 0$ is the focusing case; $p\gamma < 0$ is the defocusing case. Signs and scalings vary across papers. An equation written in laboratory time is not automatically the same experimental problem as spatial propagation of a retarded-time envelope.

Bright solitons, dark solitons on a background, breathers, and rogue-wave profiles supply useful test cases. They should not substitute for testing general random packets, interactions, changed initial conditions, or noisy observations. Higher-order, variable-coefficient, coupled, and nonlocal NLS models introduce further terms and may have different invariants and well-posedness assumptions.

For smooth periodic solutions of (2.1), standard conserved quantities are mass (or action), momentum, and Hamiltonian:

$$\begin{aligned} M[q] &= \int_{\mathbb{T}_L} |q|^2 \,\mathrm{d}y, & P[q] &= \mathrm{Im} \int_{\mathbb{T}_L} \overline{q} q_y \,\mathrm{d}y, \\ H[q] &= \int_{\mathbb{T}_L} \left(p|q_y|^2 - \frac{\gamma}{2}|q|^4 \right) \mathrm{d}y. \end{aligned} \tag{2.2}$$

These expressions assume the stated normalization and boundary conditions. On the real line, their use also requires suitable integrability. They are not a license to impose the same constraints on a different envelope model.

2.2 How a PINN is used

A conventional PINN represents $q_\theta = a_\theta + \mathrm{i}b_\theta$ by two real network outputs. Its NLS residual has real and imaginary parts

$$r_a = -\partial_s b_\theta + p\partial_{yy} a_\theta + \gamma(a_\theta^2 + b_\theta^2)a_\theta,$$
$$r_b = \partial_s a_\theta + p\partial_{yy} b_\theta + \gamma(a_\theta^2 + b_\theta^2)b_\theta.$$

Automatic differentiation supplies the local derivatives. Complex-valued networks instead use complex internal parameters or operations; both representations can express a complex solution. Their relative usefulness is an empirical question about trainability and inductive bias.

A representative objective is

$$\mathcal{L} = \lambda_r \sum_j w_j |r_\theta(z_j)|^2 + \lambda_0\mathcal{L}_0 + \lambda_b\mathcal{L}_b + \lambda_d\mathcal{L}_d + \lambda_I\mathcal{L}_I. \tag{2.3}$$

Here $z_j = (s_j, y_j)$ are residual points; $\mathcal{L}_0$ and $\mathcal{L}_b$ measure initial and boundary errors; $\mathcal{L}_d$ compares an observation operator applied to q_θ with data; and $\mathcal{L}_I$ optionally measures invariant defects. Weights require scaling and validation. A small value of this sampled, weighted objective is not itself a bound on the solution error. This basic framework has a direct NLS precedent in Raissi et al. [9].

Task	What is learned	Appropriate test
Forward solve	A field for one specified initial condition and PDE.	Error and total solve time against a converged numerical solution.
Inverse problem	Coefficients, latent initial field, or both, from observations.	Identifiability, noisy-data robustness, and withheld observations.
Assimilation	A consistent field over the observed region.	Reconstruction at withheld locations; distinguish interpolation from forecast.
Operator learning	A map from initial fields or measurements to solution fields.	Completely unseen trajectories and amortized cost including training.

A network trained on one soliton does not become a general solution operator because it can be evaluated at arbitrary coordinates. A PINO or physics-informed DeepONet is trained across inputs to learn such a map [10]. The same distinction applies to sensor-conditioned models.

2.3 The approved water-wave Dysthe equation

The current project contract specifies

$$\begin{aligned} u_\xi = & -\mathrm{i}u_{\tau\tau} - \mathrm{i}|u|^2 u \\ & -\varepsilon\left(8|u|^2 u_\tau + 2u^2\overline{u_\tau} + 2\mathrm{i}u\,|D_\tau|(|u|^2)\right). \end{aligned} \tag{2.4}$$

Here ξ is propagation distance, τ is retarded time, u is the scaled free-surface envelope, and ε is carrier steepness in the specified formal normalization. This convention agrees with the free-surface equation (1.3) of Fedele–Dutykh [11]. In the source coordinates, $\tau = \varepsilon(2x - t)$, $\xi = \varepsilon^2 x$, and $B = \varepsilon u$. Setting $\varepsilon = 0$ defines a mathematical NLS control; the physical coordinate map is not invertible at zero steepness.

On periodic profiles, the nonlocal operator is

$$\widehat{|D_\tau| f}(k) = |k|\widehat{f}(k), \qquad k = 2\pi n/L, \quad n \in \mathbb{Z}.$$

For the forward Fourier convention $e^{-\mathrm{i}k\tau}$, $|D_\tau| = \mathcal{H}\partial_\tau$; its zero mode is zero. This term depends on the full profile. Differentiating a pointwise neural output at isolated random points does not evaluate this operator.

The free-surface field is noncanonical. Use the correct transformed invariants and boundary convention if imposing an energy or momentum constraint. The potential-envelope equation, whose corresponding conjugate-derivative coefficient is zero, is a different formulation. The shared model file also requires periodic grids without a duplicated endpoint, complex arrays with axes (ξ, τ), and whole-initial-field data splits. Two array axes are not two horizontal physical directions.

3 Why a good-looking NLS solution is insufficient

There are several independent sources of failure: an inappropriate PDE; insufficient observation information; an unresolved reference calculation; a network that does not represent the relevant oscillations; and training that does not reach the relevant solution. Improving an optimizer only addresses part of this chain.

For oscillatory problems, important practical issues are spectral bias, imbalanced initial/boundary/residual gradients, large residual regions dominating small localized events, accumulated phase error, and inconsistent boundary conditions. Hard constraints, causal weighting, adaptive sampling, and better curvature information address different mechanisms [6, 12, 13, 4, 14].

Example 3.1 (Conservation can miss accumulated phase error)*.* On any periodic profile domain, consider

$$q(s,y) = e^{2\mathrm{i}s}, \qquad v(s,y) = e^{\mathrm{i}(2+\delta)s}, \qquad \delta \in \mathbb{R}.$$

The first solves (2.1) with $p = 1$, $\gamma = 2$. Both fields have the same initial data, amplitude, mass, momentum, and Hamiltonian. Yet

$$r[v] = -\delta v, \qquad \frac{\|v(s) - q(s)\|_{L^2}}{\|q(s)\|_{L^2}} = 2\,|\sin(\delta s/2)|\,.$$

Thus a small persistent residual can accumulate a large phase error. This is an exact analytic illustration, not a trained-network result.

Proposition 3.2 (A conditional NLS residual estimate)**.** *Let q, v be sufficiently smooth periodic fields on $[0, S] \times \mathbb{T}_L$, with q solving* (2.1) *and $r = \mathrm{i}v_s + pv_{yy} + \gamma|v|^2 v$. Assume p, γ are real and $|q|, |v| \le B$ everywhere. Then, for $0 \le s \le S$,*

$$\|v(s) - q(s)\|_2 \le e^{3|\gamma|B^2 s}\left(\|v(0) - q(0)\|_2 + \int_0^s \|r(t)\|_2 \,\mathrm{d}t\right). \tag{3.1}$$

Proof. Set $e = v - q$. The periodic linear Schrödinger part contributes zero to the real L^2 energy derivative. The pointwise inequality $\big||v|^2 v - |q|^2 q\big| \le 3B^2|v - q|$ and Cauchy–Schwarz give $\frac{\mathrm{d}}{\mathrm{d}s}\|e\|_2 \le 3|\gamma|B^2\|e\|_2 + \|r\|_2$ wherever the norm is nonzero. Regularization of the norm gives the same integral inequality at its zeros. Gronwall's inequality yields (3.1). □

This elementary estimate is included for explanation, not as a novel theorem. It requires a continuous residual norm, initial-error control, and amplitude bounds; a training mean-squared error does not supply these automatically. Its exponential factor can be pessimistic. It does not prove an analogous estimate for Dysthe's derivative and nonlocal nonlinearities. The more refined NLS theory discussed below uses different norms and explicit integration assumptions [7].

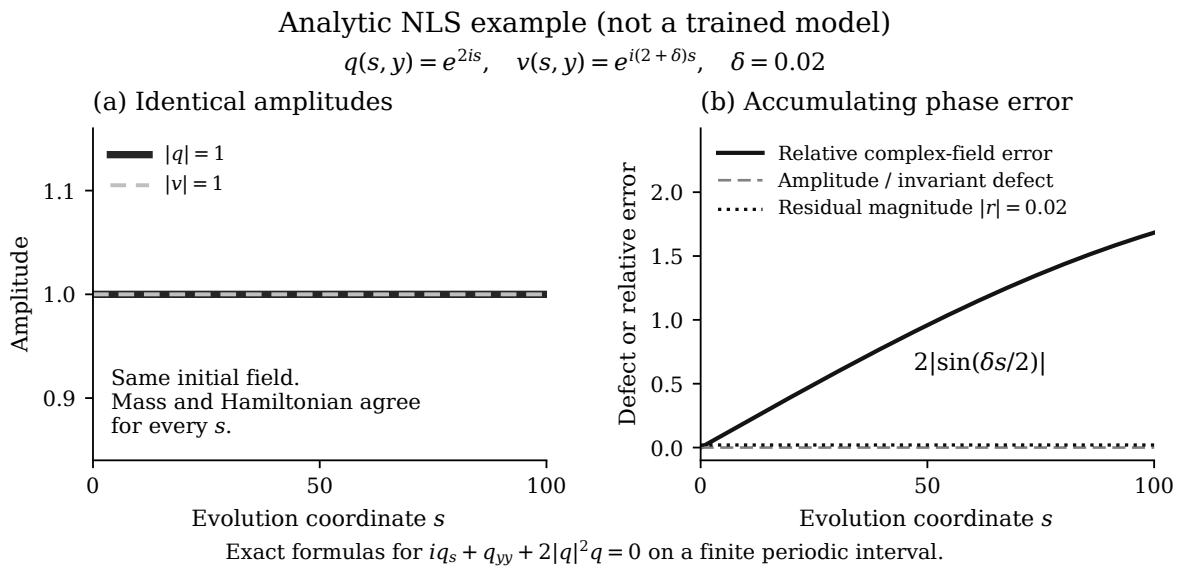


Figure 1: Exact illustration with $\delta = 0.02$. Amplitude and invariant defects are zero, while complex-field error grows over the plotted interval. The residual magnitude stays at 0.02. No PINN or Dysthe experiment is represented.

4 The strongest and newest relevant literature

4.1 Direct water-wave evidence

NLS assimilation and inverse calibration: Ehlers et al. (2024). This *Fluids* paper studies 630 synthetic higher-order-spectral (HOS) wave samples, using observations at two gauge positions and evaluating at intermediate positions. Fitting carrier-related parameters reduces reported mean surface similarity parameter (SSP) error from 0.204 to 0.186. Broader spectra and steeper waves remain difficult. The experiments use numerical data, not a laboratory validation. The paper explicitly proposes Dysthe residuals, gauge-separation studies, and actual measurements [1]. Its importance is the realistic inverse task and explicit model discrepancy. Better reconstruction with fitted parameters does not, by itself, prove that those parameters recover the true physics.

Potential-flow PINNs: Ehlers et al. (2025). Published in *Physical Review Fluids* on 2 September 2025, this study uses elevation and velocity-potential networks with potential-flow constraints. It includes laboratory flume measurements and reports a laboratory reconstruction SSP of 0.049. Forecast quality deteriorates beyond the assimilation interval; inferred laboratory velocity potential does not have independent measured ground truth. This is substantial evidence beyond analytic-soliton fitting, but uses different equations from the approved envelope model. Pretraining, faster training, and operators appear in the outlook [2].

Water-wave PINO: Ehlers, Stender and Hoffmann (2025). Published in *Physics of Fluids* on 7 October 2025, this paper already learns sparse-observation reconstruction with an FNO-based physics-informed operator. It uses 634/211/211 training/validation/test HOS samples and potential-flow constraints; reported test SSP means are 0.1035 and 0.1341 for two observation configurations. A representative 0.014 s inference time excludes training. These are synthetic, long-crested cases,

not proof of operational ocean forecasting [3]. Author code is public. A 1,056-sample HOS dataset was released on 8 April 2026 [15]. Availability was checked; neither was run here. This is the main prior-art obstacle to claiming generic sparse water-wave operator learning as new.

4.2 Direct NLS methods to reproduce or compare

Compact networks and optimization: Urbán, Stefanou and Pons (2025). The *Journal of Computational Physics* paper is a strong direct cubic-NLS baseline. Its self-scaled Broyden example uses two hidden layers of 40 units and reports real/imaginary relative errors of approximately $(5 \pm 1, 8 \pm 2) \times 10^{-6}$. Hard initial conditions and periodic features are part of the setup. Its domain differs from the original Raissi benchmark, expanding from $[-5, 5]$ to $[-15, 15]$; the nonperiodic sech tails become smaller. Consequently the cross-paper gain cannot be credited to optimization alone. The authors explicitly suggest further work on modified limited-memory BFGS [4]. Reproduce the complete setup before selecting a larger network.

Complex-valued learning: Zhang et al. (2025). The *Physical Review Research* CV-PINN paper has particularly useful accessible evidence: a full manuscript and public implementation. Complex layers and predictive dynamic monitoring are compared with an approximately parameter-matched real network across 16 one-dimensional Gross–Pitaevskii/NLS examples, with three repeats. Some training runs take approximately two GPU hours. Spectral computations supply reference fields; this does not show superiority to those solvers. Sampling improvement is not uniform in every case. The conclusion explicitly proposes richer residual/solution-gradient sampling [5]. Treat this as an architecture and training source, not as water-wave validation.

Hard-constraint ablations: Hao et al. (2026). The *Foundations of Data Science* article, early online 8 June 2026, examines the effect of initial conditions and training stability. Its NLS combination of hard initial/periodic conditions, self-adaptive weighting, and mini-batching reports real/imaginary relative errors 1.80×10^{-4} and 2.58×10^{-4}. Code is public [6]. The value lies in ablation and analysis, not the invention of hard trial functions. The findings do not imply that every noisy inverse problem should enforce measured initial data exactly. Its numbers should not be ranked against Urbán's numbers without matching the experiments.

Time decomposition and interacting structures: Chen, Li and Yuen (2025). Two direct NLS papers study interface-trained time decomposition (TDD-PINNs) and nonlinear-interaction multi-domain networks (NI-MPINNs). They include breathers, rogue-wave combinations, and inverse coefficients with synthetic noise [16, 17]. Their publisher previews report substantial gains over conventional PINNs. Full comparative budgets and code were not verified, so the gains are not independently auditable here. These are relevant precedents for domain decomposition, which should not be presented as a new idea in a Dysthe application. A spatial formulation needs propagation-distance slabs, with consistent interface fields.

Transfer between NLS cases: Chen et al. (2024). Multi-view transfer learning reuses pretrained NLS information between related initial/boundary conditions [18]. This is a useful repeated-case comparator, with publisher-level evidence inspected. Fine-tuning a pretrained solver and learning an input-conditioned operator are different algorithms. A new study must measure both adaptation cost and performance on complete unseen wave packets.

4.3 Very recent 2026 candidates

Paper/date	Method and reported evidence	Assessment and access limit
GRAVIS, 14 June [13]	Residual magnitude plus output-layer loss-gradient information; seven localized NLS cases, matched architectures/budgets; reports 1.56–2.37 times lower errors than RAD.	Strong recent sampling comparator. Publisher abstract/metadata inspected; full text, code, and complete cost audit unavailable. Parameter gradients are not the same as spatial solution gradients.
Noether-constrained PINNs, 5 September issue [19]	Dynamic conservation penalties plus adaptive residual sampling; reports small soliton amplitude error and invariant fluctuations.	Promising conservation comparator. Abstract-level access; phase, forecast horizon, repetitions, and matched costs unverified. Soft penalties do not enforce exact conservation.
CPA-PINNs, 15 October issue [20]	Phase–amplitude representation with continuity, mass and energy penalties; semiclassical two-dimensional examples and inverse tests with noise.	Newest verified phase–amplitude candidate here. Publisher preview inspected; full tables/code unavailable. Defocusing examples do not validate focusing water-wave instability.

These papers justify updating an experimental comparison. They do not justify calling one method the established best. A separate late watchlist item is Hu et al.'s *Geometry-Agnostic Data-Driven Discovery of Extreme Scenarios*, submitted 20 October 2026 [21]. Its stationary NLS/coherent-structure examples and proposed extreme-event applications are not a demonstrated Dysthe forecasting method.

Two WKB-PINN titles were screened but excluded from the confirmed-cutoff shortlist: DOI `10.1016/j.cnsns.2026.110653` has a November 2026 issue date without a verified pre-cutoff online date; the exact eligible date of DOI `10.1016/j.physd.2026.135394` was not established. The source ledger retains these uncertainties rather than assigning dates.

4.4 Transferable methods: useful, with explicit boundaries

Causality and depth. Wang, Sankaran and Perdikaris' causal PINNs weight later evolution stages according to earlier residuals [12]. This is not identical to training separate time windows. PirateNets initializes adaptive residual gates so that an effectively shallow model gains depth, and combines this with Fourier embeddings and physics-informed initialization [22]. Neither flagship paper supplied a direct NLS benchmark in the inspected material. Their application to (2.4) is a testable transfer proposal. PirateNets explicitly points toward PDE-specific embeddings and operator-learning initialization.

Optimization beyond Adam and L-BFGS. The NeurIPS 2025 gradient-alignment paper studies SOAP preconditioning on ten PDE benchmarks, reporting roughly 2–10 times better accuracy alongside approximately twice the training expense [14]. The inspected text does not establish

NLS superiority. Kiyani et al. provide broader comparisons of self-scaled quasi-Newton methods [23]. Include a compact quasi-Newton baseline before declaring that a sophisticated architecture is necessary. Compare wall time to a target accuracy, not equal optimizer iteration counts.

Operator learning and benchmark discipline. Li et al.'s PINO combines data and PDE constraints to learn across input functions [10]. That supports the many-query formulation; it is not evidence for the present Dysthe implementation. PINNacle supplies broader evidence that method strengths depend on the PDE and problem regime [24]. A shared benchmark is more informative than comparing the best numbers from unrelated papers.

Spectral and goal-oriented residuals are already prior art. Spectral informed neural networks replace differential evaluations with spectral operations [25]. Govoeyi and Richter's April 2026 preprint uses dual-weighted-residual ideas for functional-oriented sampling on Laplace problems [26]. An operator-budget study explicitly charges adaptive-sampler residual evaluations [27]. These are relevant precedents; none establishes a solution to the specific Dysthe nonlocal error-control problem proposed below. Their existence rules out claiming generic spectral PINNs or generic goal-oriented sampling as new.

4.5 Theory and classical numerical controls

NLS error bounds. Alejo, Cossetti, Fanelli, Muñoz and Valenzuela study one-dimensional subcritical focusing NLS on the real line (preprint, revised June 2025) [7]. Their result uses energy/Strichartz-type control, small defects, and integration assumptions. It is conditional error analysis, not a guarantee that gradient descent finds a suitable network. Nonzero-background breathers also need a functional setting different from directly assuming that the whole field lies in $H^1(\mathbb{R})$. This is the most directly relevant theoretical starting point found here; it does not immediately cover derivative/nonlocal Dysthe terms.

Residual certification. Eiras et al.'s ICML 2024 ∂-CROWN bounds worst-case residuals over continuous domains and includes a Schrödinger example [8]. Full text and a public code repository were inspected. A bound on a residual is not yet a bound on solution or forecast error: one still needs a stability argument and controlled initial/boundary defects. The nonlocal operator in (2.4) requires additional analysis beyond a pointwise local residual.

Do not assume PINNs beat numerical solvers. Grossmann et al.'s systematic 2024 comparison includes semilinear Schrödinger problems; the tested vanilla PINNs do not outperform FEM in total solution time and accuracy, though evaluation can sometimes be faster [28]. This does not rule out newer methods. It does establish that a neural representation alone is not a competitive numerical result. For smooth periodic envelopes, include a well-resolved Fourier-based solver and an efficient tuned solver at each required accuracy; an excessively expensive reference is not a fair speed baseline.

4.6 What is actually stated as future work?

Source	Explicit direction in inspected text	Consequence for a new paper
Ehlers 2024 [1]	Dysthe residual; gauge separation; actual observations.	A directly motivated extension, requiring a controlled contribution beyond model substitution.
Ehlers 2025 [2]	Faster training, pretraining, PINOs.	The generic operator direction is already pursued by [3].
CV-PINN [5]	Residual and solution-gradient guided resampling.	GRAVIS overlaps the broader adaptive-sampling idea; target a specific physical observable or identifiability mechanism.
Urbán et al. [4]	Modified limited-memory BFGS.	Compare memory and total solve time as well as final loss.
PirateNets [22]	PDE-specific embeddings; operator initialization.	Derive features from the approved dispersion relation and test transfer.
SOAP [14]	Less expensive preconditioned algorithms.	Require matched-accuracy cost improvements, including preconditioner work.

The five questions that follow are this review's proposals. They are not attributed to authors unless explicitly connected above. Directional-wave and alternative-formulation suggestions in the literature are not part of the current project plan.

5 Five candidate paper questions

The questions below are candidate contributions, not established novel results. A final novelty search and citation-chain review should precede submission. The recommended order is Question 1, an identifiability study from Question 2, and then the relevant method development.

5.1 Question 1: Does Dysthe improve prediction beyond fitted NLS?

Question. For which steepness, bandwidth, gauge separation, and noise regimes does the approved spatial Dysthe equation improve withheld-gauge reconstruction and downstream prediction compared with both fixed and retuned NLS?

Why this could be a paper. This directly develops the Dysthe suggestion in [1]. The scientific issue is whether adding known higher-order physics helps once a simpler model is given a fair calibration opportunity. A regime map and a causal explanation of gains or failures are stronger than a single improved error number.

Hypothesis to test. Within the justified envelope regime, Dysthe should be most useful when the leading corrections are observable but the asymptotic model still applies. It may offer little benefit at weak nonlinearity, with insufficient measurements, or outside its validity range. This is an expectation, not a demonstrated result.

Method. First use independently refined (2.4) trajectories to validate implementation. Then use separately labelled HOS-generated observations or suitable laboratory data to assess model discrepancy. Cross three model arms—fixed NLS, fitted NLS, fixed Dysthe—with two inference engines: the same tuned PINN recipe and regularized numerical assimilation around a differentiable or adjoint spectral solver. Give every arm identical observations, nuisance-parameter treatment, and tuning opportunities. Fix envelope extraction and surface reconstruction across arms.

Hold out entire initial wave realizations and entire gauge positions. Report interpolation between observed gauges separately from propagation beyond the last observed gauge. A forecast experiment must also specify which measurement times were available at the forecast origin; using a complete future time series at an upstream gauge changes the task.

Decisive result and stop condition. The target is a reproducible region in which Dysthe lowers unseen complex-field/surface-wave error or improves calibrated event predictions at a useful cost. If gains disappear under numerical assimilation, they may have come from training rather than physics. If the simpler fitted NLS performs equally well within uncertainty, publish that limit only if the benchmark and explanation are substantial; do not conceal it by selecting favorable packets.

5.2 Question 2: Is the nonlocal correction identifiable from sparse data?

Question. Can the observations distinguish mean-flow effects from carrier calibration, the unknown inlet field, and local nonlinear terms? Which sensor arrangement or wave-packet family makes this distinction possible?

Why this could be a paper. Predictive success does not imply parameter recovery. A parameter may absorb model discrepancy or compensate for an uncertain initial field. The contribution would be an identifiability limit and a practical observation-design remedy. This is our proposed extension of the inverse problem, not a published Dysthe identifiability theorem.

Method. Before training a neural network, form the observation map $d \approx \mathcal{O}(\alpha, z_0)$ using the verified solver, where α denotes diagnostic coefficients and nuisance calibration parameters, and z_0 denotes an inlet parametrization. Compute scaled sensitivities

$$J_\alpha = \frac{\partial \mathcal{O}}{\partial \alpha}, \qquad J_0 = \frac{\partial \mathcal{O}}{\partial z_0}.$$

Study the singular values of the joint, noise-whitened Jacobian and the information remaining for α after accounting for z_0. A small singular value signals local confounding; full rank alone does not prove global identifiability. Supplement it with profile likelihoods, multi-start estimation, repeated-noise trials, and confidence-interval coverage for known synthetic parameters.

Compare two gauges with three or more, altered separation, and deliberately varied packet bandwidth and phase modulation. A sensor-selection criterion can maximize the smallest relevant singular value or a regularized information determinant. Compare it with uniform placement and a simple greedy heuristic. Afterwards test whether the same information gains improve PINN and numerical assimilation.

Model identity and decisive result. The approved physical baseline keeps its coefficient tuple fixed. Coefficient fitting, removing the nonlocal term, or scaling individual terms creates separately named diagnostic models, never relabelled baseline evidence. A useful outcome is an experimentally

testable minimum information condition. If observations cannot distinguish the terms, report predictive uncertainty rather than a precise inferred coefficient.

5.3 Question 3: Which intervention prevents long-distance phase failure?

Question. Can propagation-wise training and a correctly evaluated nonlocal residual maintain phase and focusing-event accuracy at a lower total cost than conservation penalties or residual adaptivity alone?

Novelty boundary. Neither conservation-aware NLS PINNs, spectral differentiation, time decomposition, nor goal-oriented sampling is new [19, 20, 16, 25, 26]. The prospective contribution is a demonstrated mechanism linking Dysthe's nonlocal coupling, accumulated phase error, and measurable focusing-event failures.

Method. Use structured full-τ slices so that the mean-flow term is evaluated by a differentiable Fourier multiplier with controlled padding and quadrature. Compare local automatic differentiation plus the common nonlocal operator with a fully spectral profile residual. An optional weak formulation integrates local derivatives by parts and moves the self-adjoint multiplier onto a suitable test function where justified. Check weak residuals against an independent strong residual, since an incomplete test basis can miss modes.

Start with a compact real-output network and a parameter-matched complex network. Compare global training, causal weights in ξ, and sequential ξ slabs; distinguish these interventions. Then compare uniform, RAD, and sensitivity/goal-oriented sampling with equal total budgets, including candidate scoring and interface work. For a smooth proxy to a maximum, one possible functional is

$$J_\kappa(u) = \kappa^{-1} \log\left(\frac{1}{|\Omega|}\int_\Omega e^{\kappa|u|^2}\,\mathrm{d}\xi\,\mathrm{d}\tau\right), \qquad \kappa > 0.$$

Finite κ introduces a deliberate approximation; validate the actual peak height and location afterwards. Do not tune κ on the final test events.

Decisive result and risk. Require better unaligned phase/event error versus wall time, not only lower invariant drift. Independently vary grid, domain, and residual quadrature. If a benefit vanishes when aliasing or sampler overhead is counted, it is not a method improvement. A narrow ablation within Question 1 may be more valuable than a separate architecture paper.

5.4 Question 4: When does an operator pay for its training?

Question. Can an input-conditioned Dysthe surrogate reliably generalize to unseen wave packets and achieve a matched-accuracy cost advantage over repeated numerical solves or inverse fits?

Novelty boundary. Generic water-wave PINO reconstruction already exists [3]. Specify whether the new operator maps initial fields to trajectories or measurements to reconstructed fields. These are different datasets and claims. The prospective advance is phase-aware generalization, controlled physics correction, and calibrated failure detection for the approved model.

Method. Compare a supervised FNO, a physics-informed operator using the exact residual, and a hybrid that retains an accurate linear propagator and learns a constrained nonlinear correction. Add a per-instance PINN, pretrained-and-refined PINN, and numerical solver/assimilation controls.

Split whole initial fields, including all refinements and augmentations. Reserve distinct bandwidth, steepness, and sensor configurations for out-of-distribution testing; do not mix them into a single average.

For an operator with total preparation cost C_{prep}, inference plus adaptation cost C_{op}, and a classical per-query cost C_{ref}, the simplest break-even estimate is

$$Q_\star = \frac{C_{\mathrm{prep}}}{C_{\mathrm{ref}} - C_{\mathrm{op}}}, \qquad C_{\mathrm{ref}} > C_{\mathrm{op}}.$$

Round upward for integer query counts. If the denominator is nonpositive, there is no speed break-even under this model. Charge data generation, preprocessing, tuning, training, failed runs, and deployment-specific refinement consistently. Compare solutions at the same target accuracy.

Decisive result and risk. Demonstrate both a useful $Q_\star$ and acceptable held-out phase/event errors, with calibrated abstention when the input leaves the validated regime. Ensemble disagreement alone is not calibrated uncertainty: assess coverage and failure-detection rates on separate trajectories. A fast model that misses rare focusing events fails the intended task.

5.5 Question 5: Can a Dysthe prediction have a useful error certificate?

Question. Can one compute a finite-propagation bound on the error of a learned Dysthe trajectory that is tight enough to remain informative through nonlinear focusing?

Novelty boundary. NLS stability-based bounds and continuous-domain PINN residual certification already exist [7, 8]. Repeating a generic Gronwall argument is not a new contribution. The challenge is a computable, nonvacuous bound for the actual derivative/nonlocal formulation.

Method. Begin with the finite-dimensional Fourier discretization $\dot{z} = F_N(z)$. Bound the discrepancy of a learned trajectory using its initial defect and an independently bounded residual. Investigate interval- or energy-based local stability estimates, possibly per propagation slab, rather than an unnecessarily large global Lipschitz constant. Quantify certificate overhead and effectivity (bound divided by observed error on resolved test cases).

A certificate relative to this ODE is not yet a certificate for the continuum PDE. A second stage must control Fourier truncation, quadrature, aliasing, regularity, and the chosen periodic or whole-line setting. Physical model discrepancy and observation uncertainty are additional errors, not consequences of the numerical certificate. State explicitly which of these quantities is bounded rigorously and which is only measured.

Decisive result and risk. The target is a bound useful before and after representative focusing events, with tractable cost. If all rigorous bounds become vacuous long before those events, a shorter-horizon result may still be valuable, but an empirical uncertainty band must not be renamed a proof. This is the highest-risk direction and would benefit from PDE/numerical-analysis supervision.

6 A concrete experimental protocol

6.1 Establish three separate levels of evidence

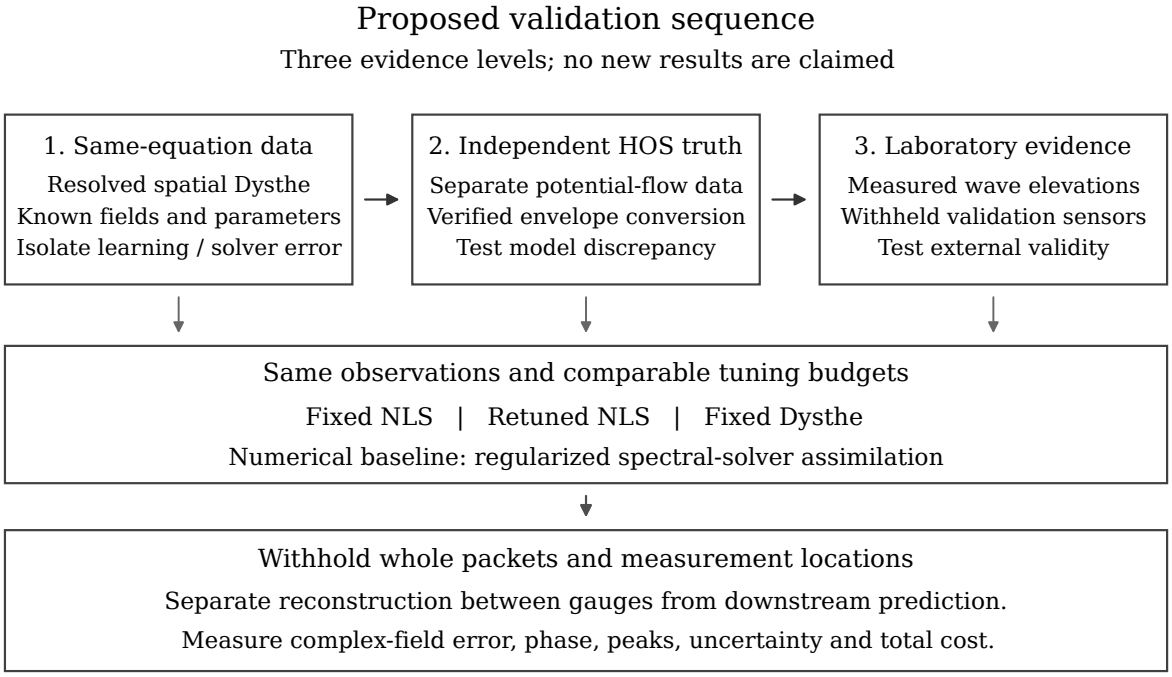


Figure 2: Proposed study design. Same-equation synthetic tests establish implementation accuracy; independently generated HOS and laboratory data address different model-validity questions. These levels must keep their original physical provenance. The diagram contains no experimental results.

1. **Numerical accuracy for the declared equation.** Use plane waves and the mathematical NLS control for elementary checks, then converged Dysthe packet trajectories. Independently refine propagation step, profile resolution, nonlinear quadrature, and domain length. Agreement with one's own unresolved training data is insufficient.
2. **Approximation of more complete water-wave physics.** Use HOS truth with its original model label and a documented envelope extraction map. The public dataset [15] is a candidate source, not automatically a Dysthe benchmark. Select compatible unidirectional, deep-water, narrowband cases and justify the validity bounds.
3. **Measurement performance.** Use laboratory observations with calibration and noise information. Assess only quantities independently observed or otherwise supported. Reconstructing an unmeasured latent potential is not its experimental validation. Do not make operational sea-state claims from tank data alone.

6.2 Minimal pilot before a large campaign

The following is a proposed workload, not an existing result or a universal sample-size requirement. Begin with three regimes spanning low to moderate steepness/bandwidth within justified limits, two

gauge separations, two noise levels, and five independently drawn packets per cell: $3 \times 2 \times 2 \times 5 = 60$ inverse cases. The noise-free level checks implementation; the other should reflect an explicit sensor-noise model. Choose physical parameter values only after reference-resolution and envelope-validity checks. The 60 cases are for feasibility, not enough by themselves to settle all regime interactions.

Run the three equation arms and two inference engines of Question 1. Start with a few optimizer seeds on the pilot, then choose replication from observed variability. For the final comparison, use paired initial fields and noise realizations, report median and upper-tail errors, and bootstrap whole trajectories rather than individual grid points. Record optimization failures; do not discard them from the success rate. A larger study can use a space-filling design followed by targeted sampling near changes in the relative model performance.

6.3 A practical PINN starting recipe

The following is a proposed baseline assembled from inspected methods; its Dysthe performance remains to be established.

1. **Scale the problem.** Use the recorded physical normalization, nondimensional coordinates, and training-only feature scaling. Start with double precision for reference and residual qualification, then measure whether reduced precision preserves the required accuracy.
2. **Represent periodicity correctly.** For an exact periodic inlet u_0, use commensurate features such as $\cos(2\pi\tau/L), \sin(2\pi\tau/L)$ and an ansatz

$$u_\theta(\xi, \tau) = u_0(\tau) + \frac{\xi}{\Xi} N_\theta\left(\frac{\xi}{\Xi}, \cos(2\pi\tau/L), \sin(2\pi\tau/L)\right), \quad 0 \le \xi \le \Xi.$$

   This enforces the initial field and periodic profile by construction. If u_0 is uncertain, infer a latent inlet and penalize its disagreement with measurements according to the noise model instead of hard-coding noisy observations. Random Fourier frequencies are not automatically periodic with period L.
3. **Evaluate the nonlocal term on complete slices.** Evaluate the field on a uniform periodic profile grid at sampled ξ values, apply a differentiable FFT multiplier, and evaluate products using justified anti-aliasing/projection choices. Validate the result independently on a finer grid. Record that the residual contains a numerical spectral operator.
4. **Use a credible optimizer comparison.** Compare an Adam-to-L-BFGS recipe with self-scaled Broyden and, if resources permit, SOAP. Resampling changes the optimization objective and may interact with quasi-Newton history; specify when points are frozen or history is reset. Count line searches and derivative evaluations.
5. **Add complexity one ingredient at a time.** After the basic model works, test ξ slabs/causal weights, complex layers, adaptive sampling, and appropriate invariant penalties. Tune each on validation cases. A large combination without ablations does not explain why a method works.

6.4 Metrics that match the scientific claim

Let u denote a resolved reference and $\widehat{u}$ a prediction on a fixed evaluation domain Ω, with physical quadrature weights. Report at least

$$E_{\text{field}} = \frac{\|\widehat{u} - u\|_{L^2(\Omega)}}{\|u\|_{L^2(\Omega)}}, \qquad E_{\text{amp}} = \frac{\||\widehat{u}| - |u|\|_{L^2(\Omega)}}{\|u\|_{L^2(\Omega)}}. \tag{6.1}$$

For meaningful phase evaluation define $A_\eta = \{(\xi, \tau) : |u| \geq \eta,\ |\widehat{u}| \geq \eta\}$ using a documented amplitude threshold and use, for example,

$$E_\phi^2 = \frac{\int_{A_\eta} |u|^2 \operatorname{Arg}(\widehat{u}\,\overline{u})^2 \,\mathrm{d}\xi\,\mathrm{d}\tau}{\int_{A_\eta} |u|^2 \,\mathrm{d}\xi\,\mathrm{d}\tau}.$$

If the denominator vanishes, report the metric as undefined. Also report the fraction of the domain retained by the threshold. Phase- or translation-aligned errors can diagnose mistakes, but must be accompanied by unaligned errors and the fitted shifts.

Additional measurements should include:

- envelope peak height, focusing distance, retarded-time location, and event detection precision/recall at preregistered thresholds;
- profile spectra and spectral-edge content; correct invariant defects, interpreted alongside reference discretization error;
- withheld surface-elevation error after a fixed observation map; SSP may aid comparison with water-wave papers but is not interchangeable with complex-field relative L^2 error;
- coefficient recovery and calibrated interval coverage when parameters are actually identifiable;
- training, tuning, data-generation, adaptation and inference time, peak memory, and total residual/operator evaluations.

Never label a small invariant defect, a low training loss, or a visually plausible envelope an accuracy certificate.

6.5 Fair comparisons and explicit decision gates

Separate a best-accuracy experiment from a fixed-budget experiment. Allow comparable hyperparameter-selection resources, keep the final test set untouched, and disclose hardware and numerical precision. Charge adaptive candidate scoring and sampler maintenance [27]; iteration counts alone can hide significant costs.

Before the final campaign, choose a minimum practically useful improvement and tolerable cost ratio. For example, a *proposed* pilot continuation rule could require a 20% reduction in median held-out phase or field error, a paired confidence interval excluding no improvement, and no material degradation in upper-tail event error. This number is a project decision to be justified, not a literature standard. If the reference uncertainty is comparable to the claimed improvement, tighten the reference first. If no practically useful gain appears, simplify the method or redirect to the documented identifiability limitation.

7 Roadmap, current assets, and delivery limits

7.1 A staged route to a submission

The schedule below is an indicative planning estimate, conditional on compute, data access, and successful validation.

Stage	Deliverable	Gate before proceeding
Weeks 1–2	Reproduce one direct NLS baseline; verify the exact Dysthe residual and observation conversion.	Correct signs, mean flow, boundary conditions, and independent refinement evidence.
Weeks 3–4	Numerical assimilation and modern PINN pilot; local sensitivity/identifiability analysis.	Equal observations and credible simpler-model controls; meaningful effect size.
Weeks 5–7	Expand regimes, seeds, and held-out packets; one targeted ablation from Question 3.	Stable gains or a robust, explainable failure boundary.
Weeks 8–10	External HOS or laboratory validation; uncertainty and cost evaluation; manuscript and reproducibility package.	Claims match the actual evidence level and survive negative controls.
Later	Operator amortization or rigorous certification.	Strong first-stage baselines; separate novelty and feasibility review.

Do not begin all five questions simultaneously. Question 1 with a focused part of Question 2 supplies a coherent first manuscript. Question 3 can provide a method explanation if the pilot identifies a genuine training bottleneck. Question 4 becomes sensible only when repeated-query demand and dataset costs are known; Question 5 is a separate mathematical track.

7.2 What is available locally now

Read-only inspection of the current family confirmed the model contract and an implemented `SpatialDysthe` reference solver in `dysthe-core/src/dysthe_core/reference/spatial.py`. The equation specification is `dysthe-core/docs/model.md`; qualification guidance is `dysthe-core/docs/stage-one-reference.md`. The FNO and hybrid method READMEs in `dysthe-learning` still describe planned methods. This is a current file inspection, not a fresh numerical qualification or training result. Historical artifacts must retain their actual equation labels.

The current solver is therefore a starting asset to qualify, rather than a reason to build an unrelated simulator. The paper should use `dysthe-experiments` for frozen campaigns and splits, `dysthe-learning` for compared methods, and `dysthe-paper` for numerical evidence once experiments exist. This review does not modify those implementations or assert that the proposed experiments have been completed.

7.3 Verification and limitations of this review

This report provides live primary-source literature checking, access and status distinctions, explicit future-work provenance, an independent check of the elementary NLS example and estimate, and original vector figures. Several very recent Elsevier/Springer papers were accessible only through

abstracts and previews; their full comparative claims, code, and statistical evidence could not be audited. The stronger full-text sources and those preview-only leads are identified separately.

No literature experiment, public training code, HOS dataset, or local solver qualification campaign was rerun. No definitive claim of novelty, universal best method, ocean-operational readiness, or established Dysthe–PINN advantage is made. The main deliverable is a concrete, testable research program grounded in the verified sources below.

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